Bhavana Sudhakar Athavane

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Education

California State University, Fullerton — Fullerton, CA
Master of Science, Computer Science — *May 2026*

Jyothy Institute of Technology — Bangalore, India
Bachelor of Engineering, Computer Science — *July 2022*

Experience

California State University — Fullerton, CA
Graduate Teaching Assistant — *January 2026 – August 2026*

- Instructed SQL and applied analytics to 50+ students/semester; redesigned lab exercises around joins, aggregations, window functions, query debugging, and techniques for handling large datasets (100K+ rows) in Microsoft Excel

Nokia — Sunnyvale, CA
*Platform Engineering Architect → Database Software Engineer (**Co-Op**)* — *May 2025 – December 2025*

- Built an incident-driven root-cause diagnostic PoC on Google Cloud Vertex AI, combining generative models with retrieval augmented (RAG) search over historical incident data; tuned prompts and validated recommendations against real Jira incidents, achieving ∼82% root-cause match accuracy and cutting MTTR by ∼41%
- Architected a GCP data pipeline (Pub/Sub → Dataflow → BigQuery) to ingest and transform incident log bundles and diagnostic data from 200+ network nodes; reduced manual triage effort by ∼80% and enabled automated alerting
- Scaled the retrieval layer (RAG) using Vertex AI Search (Discovery Engine), unifying CMDB, Jira, and distributed knowledge bases into a single cross-source semantic search backend; improved runbook relevance and reduced incident resolution time by ∼35%
- Designed and optimized 4+ relational database schemas across Oracle, MySQL, and PostgreSQL underpinning the platform's AI-driven backend systems; reduced average query latency by ∼55% through query-plan analysis, SQL rewrites, and indexing
- Built Flask-based REST API integrations connecting backend services to OLTP-like data sources (Jira, CMDB), standardizing data access and reducing redundant database queries

Cognizant — Bangalore, India
Senior Systems Engineer — *March 2022 – August 2024*

- Led an Oracle DB performance engineering initiative across 3 production systems, improving overall throughput by 30% through query-plan analysis, index rebuilding, and partition pruning; directly supporting SLA compliance
- Designed and scaled an automated SQL data-validation framework in Python and PL/SQL spanning Oracle DB and SQL Server across 4+ production pipelines; caught ∼20 cross-system reporting discrepancies per month
- Built and delivered 6+ executive Tableau dashboards tracking KPIs across revenue, operations, and compliance, enabling data-driven decision-making for senior stakeholders
- Automated over 8+ internal ETL workflows using Python and SQL, reducing end-to-end processing time by ∼15%

Technical Skills

- **Languages & Platforms:** Python, SQL, Java, JavaScript, Git, Tableau, Jira, REST APIs, Flask
- **Cloud & AI/ML:** GCP, Vertex AI, BigQuery ML, Cloud Storage, AWS Redshift, LLM Integration, RAG, Semantic Search, NLP
- **Data Engineering:** ETL Pipelines, Database Design, Query Optimization, Stored Procedures, Pub/Sub, Dataflow
- **Databases:** Oracle DB, MySQL, SQL Server, PostgreSQL

Projects

AI-Powered Financial Fraud Detection System

- Built an end-to-end financial fraud detection pipeline covering transaction ingestion, feature engineering, and anomaly classification; model identified high-risk patterns across a dataset of 1M+ synthetic transactions with 89% precision
- Engineered a feature set from transaction metadata (velocity, geolocation deltas, merchant category patterns) to improve model separability between fraudulent and legitimate transactions
- Evaluated model performance using precision-recall tradeoffs suited to class-imbalanced fraud data, tuning classification thresholds to minimize false positives while preserving recall on rare fraud cases

Content Modeling using NLP

- Developed an NLP document processing pipeline using CNN models and OCR to extract and structure unstructured text from 5+ document formats, reducing manual categorization effort for a 10K-document test corpus
- Built a preprocessing layer for OCR noise correction and layout normalization, improving downstream text extraction accuracy across scanned and image-based document formats
- Designed a document classification scheme to auto-tag extracted content by category, structuring previously unstructured data for downstream search and retrieval